

RESEARCH ARTICLE

Quantifying Effect Modifier Similarity for Regional Pooling in Multi-Regional Clinical Trials

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Abstract

The ICH E17 guideline recommends regional pooling in multi-regional clinical trials (MRCTs) based on similarity of effect modifier distributions, but provides no quantitative methodology. We propose the normalized Area Between Cumulative Distributions (nABCD), a planning-stage dissimilarity index defined as the Wasserstein-1 distance divided by twice the pooled interquartile range. Unlike the standardized mean difference (SMD), nABCD captures differences in variance, shape, and skewness—distributional features that may drive treatment effect heterogeneity through non-linear effect modification. Because nABCD requires only baseline effect modifier distributions, it can be computed from any representative data source, including prior trials, disease registries, and real-world evidence databases—enabling a common planning scenario in which a small-sample region identifies suitable pooling partners before the trial begins. When prior evidence on conditional average treatment effect (CATE) sensitivity is available, clinical calibration translates nABCD into maximum potential regional treatment effect differences ($\Delta_{\max}$) with bootstrap confidence intervals. Simulation studies across eight scenarios demonstrated satisfactory bias and coverage of the percentile bootstrap estimator for non-negligible distributional differences ($\text{nABCD} \geq 0.1$) at $n \geq 100$ per region. Application to two publicly available stroke trial datasets illustrates both common planning scenarios: Case A (effect modifier unknown; IST-1, 31 countries) using sensitivity analysis, and Case B (effect modifier identified; IST-3, 8 countries) using clinical calibration. The IST-3 results demonstrate that similar distributional magnitudes can yield markedly different clinical conclusions: NIHSS ($\text{nABCD} = 0.240$, $\Delta_{\max} = 5.02$ percentage points) versus age ($\text{nABCD} = 0.285$, $\Delta_{\max} = 0.47$ percentage points), underscoring that clinical calibration—not nABCD magnitude alone—should guide pooling decisions. The nABCD framework fills a methodological gap in ICH E17 implementation, supporting evidence-based pooling decisions from any representative population data source.

KEYWORDS

multi-regional clinical trial; ICH E17; effect modifier; Wasserstein distance; regional pooling; distributional similarity

1 | INTRODUCTION

Multi-regional clinical trials (MRCTs), conducted across multiple countries or regulatory regions under a single protocol, have become the standard paradigm for global pharmaceutical development.^{1,2} This approach offers substantial benefits: accelerated timelines, broader generalizability, and earlier access to new therapies for patients worldwide. The International Council for Harmonisation (ICH) E17 guideline, adopted in 2017, established principles for planning and designing MRCTs, with a central assumption that treatment effects are generalizable across the target population.³

Several regulatory authorities expect demonstration of treatment effect consistency in regional subpopulations,^{4,5} yet individual regional sample sizes in global MRCTs are often insufficient for such demonstration. A key strategy for addressing this challenge

Abbreviations: ABCD, area between cumulative distributions; CATE, conditional average treatment effect; CDF, cumulative distribution function; CI, confidence interval; EU, European Union; ICH, International Council for Harmonisation; IPD, individual patient data; IQR, interquartile range; IST-3, Third International Stroke Trial; KL, Kullback–Leibler; KS, Kolmogorov–Smirnov; MRCT, multi-regional clinical trial; nABCD, normalized area between cumulative distributions; NIHSS, National Institutes of Health Stroke Scale; NMPA, National Medical Products Administration; OHS, Oxford Handicap Scale; SMD, standardized mean difference; US, United States.

is the *regional pooling approach*, wherein regions with similar patient characteristics are grouped for analysis to enhance the ability to assess regional consistency.³ The ICH E17 guideline describes pooling decisions in terms of the similarity of effect modifier distributions.³

An effect modifier is a baseline patient characteristic—such as age, disease severity, or genetic marker—for which the treatment benefit differs across subgroups. For example, if younger patients respond better to treatment than older patients, age is an effect modifier. When such heterogeneity exists, even if the drug works identically at the individual level, regions with different patient compositions may observe different average treatment effects. A region with predominantly younger patients would show larger benefits than a region with predominantly older patients, not because the drug works differently, but because the patient mix differs. This fundamental relationship underscores why effect modifier distributional similarity is critical to the validity of regional pooling.

Despite the regulatory importance of effect modifier distributional similarity, current practice lacks a standardized quantitative methodology. The ICH E17 guideline provides no specific metric, threshold, or statistical procedure for determining when distributions are “similar enough.” Recent regulatory guidance has highlighted this gap. Song et al., writing from the China NMPA perspective on ICH E17 implementation, note the challenge of operationalizing pooling criteria without quantitative tools.⁵ Long et al. further discuss basic considerations for consistency evaluation under ICH E17.⁶

This absence of quantitative tools is particularly acute at the *planning stage*, when sponsors must identify suitable pooling partners for small-sample regions. The pooling rationale requires quantitative evidence that partner regions’ patient populations are “similar enough,” yet no established method exists to provide such evidence. A workshop report by Matsushima et al. documented that regional imbalance in CRP-positive/MRI-negative status contributed to apparent inconsistency in the secukinumab MRCT.⁴ This example illustrates the practical consequences when effect modifier distributional differences are not assessed quantitatively at the planning stage.

Although general-purpose comparison tools such as the standardized mean difference are routinely applied to baseline covariates,⁷ they were not designed for assessing effect modifier distributional similarity across regions. As a single summary measure, the standardized mean difference cannot detect differences in variance or distributional shape. When effect modification is non-linear, these distributional features directly influence regional average treatment effects, and a comparison based on means alone will miss them.

This paper addresses these gaps by proposing the *normalized Area Between Cumulative Distributions* (nABCD) as a quantitative index for assessing effect modifier distributional similarity across regions. nABCD is defined as the Wasserstein-1 distance between two cumulative distribution functions, normalized by a robust scale measure (the pooled interquartile range) to achieve scale-free interpretation. Unlike the standardized mean difference, nABCD captures differences in variance and shape simultaneously. We estimate nABCD with bootstrap confidence intervals and link it to clinical interpretation through a calibration framework. We focus on continuous effect modifiers, for which the Wasserstein distance provides a theoretically grounded measure of distributional dissimilarity.

The remainder of this paper is organized as follows. Section 2 presents the methodological framework, including the index definition, clinical calibration, and estimation procedure. Section 3 describes a simulation study. Section 4 demonstrates the framework using publicly available individual patient data. Section 5 discusses implications, limitations, and future directions.

2 | METHODS

An effect modifier is a factor for which the treatment effect varies across subgroups. Formally, let the conditional average treatment effect (CATE) be denoted $\tau(x) = E[Y(1) - Y(0) \mid X = x]$, where X is the effect modifier. When this function is non-constant, the average treatment effect observed in region r depends on the distribution F_r of the effect modifier in that region:

$$\bar{\tau}_r = \int \tau(x) dF_r(x). \quad (1)$$

Assessing pooling suitability therefore requires a measure of distributional difference between regions.

2.1 | Existing Approaches and Their Limitations

The most widely used measure for comparing covariate distributions across groups is the standardized mean difference (SMD), defined as

$$\text{SMD} = \frac{\bar{X}_1 - \bar{X}_2}{S_{\text{pooled}}}, \quad (2)$$

where $\bar{X}_r$ denotes the sample mean in region r and S_{pooled} is the pooled standard deviation.⁷ Because treatment effects in clinical trials are typically summarized as mean differences, SMD provides a scale-free comparison directly relevant to the quantities that drive clinical decision-making. However, by construction SMD captures only differences in location: two distributions with identical means but markedly different variances—for example, $N(50, 5^2)$ and $N(50, 15^2)$ —yield $\text{SMD} = 0$ despite a threefold difference in spread.

Distributional distance measures address this limitation by comparing distributions beyond summary statistics. Empirical distribution function (EDF) tests such as the Kolmogorov–Smirnov (KS) statistic, $D_{\text{KS}} = \sup_x |F_1(x) - F_2(x)|$, and related measures (Cramér–von Mises, Anderson–Darling) quantify differences in the full shape of distributions by comparing CDFs directly. Information-theoretic divergences such as Kullback–Leibler (KL) divergence,

$$D_{\text{KL}}(P||Q) = \int p(x) \log \frac{p(x)}{q(x)} dx, \quad (3)$$

take a density-based approach, measuring how one probability distribution differs from another.

However, each of these approaches has limitations when applied to comparing effect modifier distributions across regions. SMD captures only location differences and is blind to variance and shape. EDF-based statistics such as the KS statistic capture full distributional differences but lack a clinically interpretable scale and have no theoretical connection to treatment effect heterogeneity. KL divergence is asymmetric—comparing region A to region B yields a different value than comparing B to A—can diverge to infinity when empirical distributions do not fully overlap, and requires density estimation that is impractical with small regional samples. Crucially, none of these measures offers a theoretical link that would translate a distributional distance into a bound on regional treatment effect differences. These gaps motivate a measure that (i) captures distributional features beyond location, (ii) provides a clinically interpretable scale, and (iii) offers such a theoretical link. The nABCD index, defined in the following subsection, is designed to satisfy all three requirements.

2.2 | The nABCD Dissimilarity Index

Among distributional distances that could satisfy these three requirements, we adopt the Wasserstein-1 distance (W_1) for its unique theoretical connection to treatment effect heterogeneity (Proposition 2 below). The W_1 distance (sometimes called the Earth Mover’s Distance) between two cumulative distribution functions F and G is defined as:⁸

$$W_1(F, G) = \int_{-\infty}^{\infty} |F(x) - G(x)| dx. \quad (4)$$

Geometrically, this equals the total area between the two CDFs (Figure 1). Unlike the standardized mean difference, the Wasserstein distance responds to changes in variance, skewness, and other distributional features.⁹

The *normalized Area Between Cumulative Distributions* (nABCD) is defined as the Wasserstein-1 distance normalized by twice the pooled interquartile range:

$$\text{nABCD}(F_1, F_2) = \frac{W_1(F_1, F_2)}{2 \cdot \text{IQR}_{\text{pooled}}}, \quad (5)$$

where $\text{IQR}_{\text{pooled}} = Q_{0.75}(F_{\text{pool}}) - Q_{0.25}(F_{\text{pool}})$ with $F_{\text{pool}} = \lambda F_1 + (1 - \lambda)F_2$ and $\lambda = n_1/(n_1 + n_2)$. The denominator normalizes W_1 by a scale estimator to achieve scale-free interpretation. We adopt the interquartile range (IQR) for its direct interpretability as the central 50% spread of the effect modifier distribution. The factor of 2 calibrates nABCD so that a one-IQR location shift yields $\text{nABCD} = 0.5$.

W_1 is a proper distance satisfying the triangle inequality.⁸ Normalization by the pair-dependent pooled IQR does not preserve this property, but it enables comparison of distributional similarity across effect modifiers measured on different scales—a practical necessity when assessing multiple covariates simultaneously. nABCD thus serves as a dissimilarity index retaining non-negativity, symmetry, and identity of indiscernibles (Proposition 1).

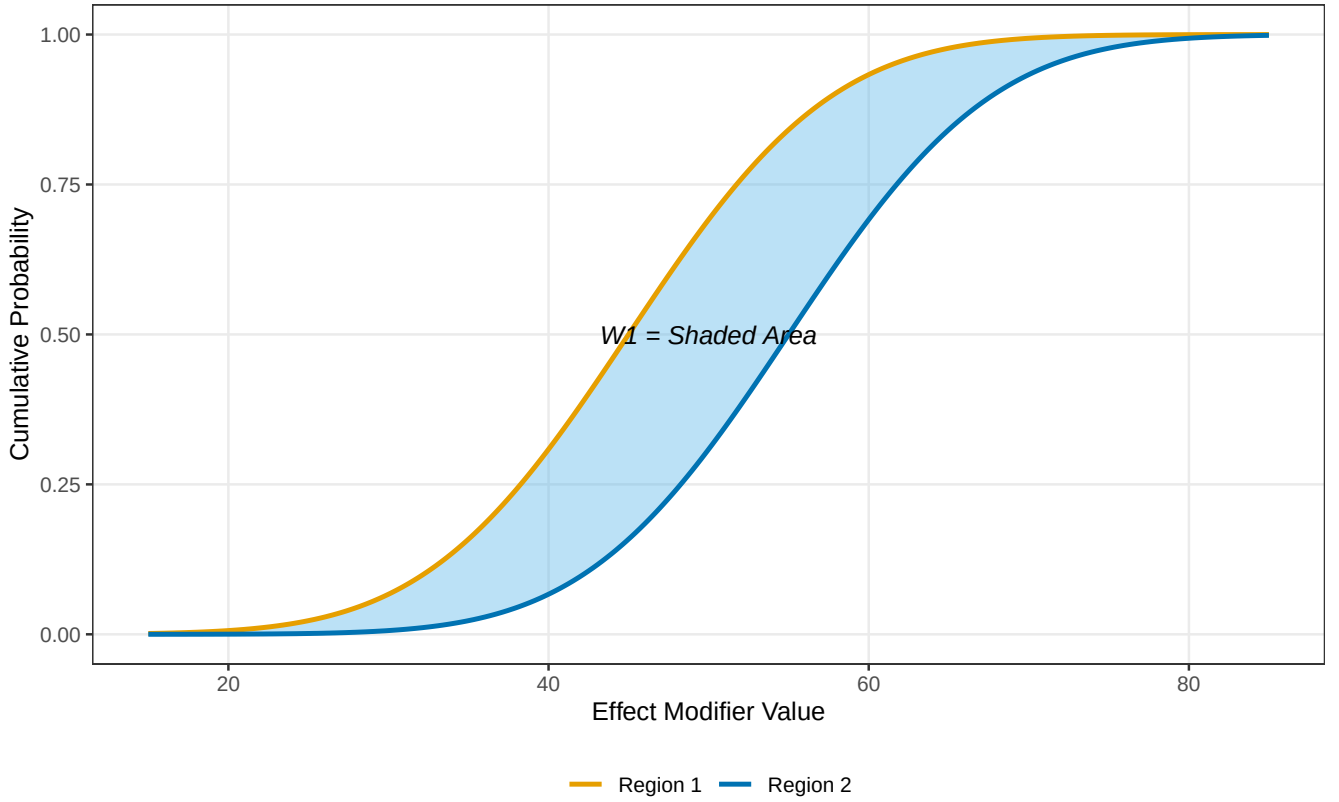


FIGURE 1 nABCD as the area between cumulative distribution functions. The shaded region represents the Wasserstein-1 distance $W_1(F_1, F_2)$, which equals the total area between the two CDFs. nABCD normalizes this area by twice the pooled IQR to achieve scale-free interpretation.

Proposition 1. (Non-negativity). *For distributions with finite first moment, provided $\text{IQR}_{\text{pooled}} > 0$ (i.e., the pooled distribution is non-degenerate), $nABCD \geq 0$, with equality if and only if $F_1 = F_2$.*

The specific choice of W_1 as the underlying distance is motivated by the Kantorovich–Rubinstein duality theorem,⁸ which provides a theoretical bridge between distributional distance and treatment effect heterogeneity. The connection proceeds in three steps:

1. *W_1 as CDF area:* In one dimension, $W_1(F_1, F_2) = \int |F_1(x) - F_2(x)| dx$ (equation 4), the total area between two CDFs. This area increases whenever the distributions differ in location, spread, or shape.
2. *Kantorovich–Rubinstein duality:* The W_1 distance has an equivalent dual representation:⁸

$$W_1(F_1, F_2) = \sup_{\|f\|_{\text{Lip}} \leq 1} \left| \int f dF_1 - \int f dF_2 \right|.$$

That is, W_1 equals the largest possible difference in expected values of f between the two distributions, where the supremum is taken over all functions whose rate of change is bounded by 1 ($|f(x) - f(y)| \leq |x - y|$ for all x, y ; such functions are called 1-Lipschitz).

3. *CATE application:* A function is Lipschitz continuous with constant L if its rate of change is bounded by L : $|\tau(x) - \tau(y)| \leq L|x - y|$ for all x, y . In clinical terms, L controls how much the treatment effect can change per unit change in the effect modifier. Since τ/L is 1-Lipschitz, applying the duality yields:

$$|\bar{\tau}_1 - \bar{\tau}_2| \leq L \cdot W_1(F_1, F_2). \quad (6)$$

This Lipschitz-dual characterization is specific to W_1 . The W_2 distance, while admitting closed-form expressions for Gaussian families, does not possess it and therefore cannot provide such a bound. The alternative measures discussed in Section 2.1 similarly lack any analogous property.¹⁰

Combining the bound above with the nABCD definition (equation 5) yields the central result:

Proposition 2. (Connection to heterogeneity). *If the CATE function is Lipschitz continuous with constant L , then*

$$|\bar{\tau}_1 - \bar{\tau}_2| \leq 2L \cdot \text{IQR}_{\text{pooled}} \cdot \text{nABCD}(F_1, F_2). \quad (7)$$

Proof. Substituting (5) into (6) yields $|\bar{\tau}_1 - \bar{\tau}_2| \leq L \cdot W_1(F_1, F_2) = L \cdot 2 \cdot \text{IQR}_{\text{pooled}} \cdot \text{nABCD}(F_1, F_2) = 2L \cdot \text{IQR}_{\text{pooled}} \cdot \text{nABCD}(F_1, F_2)$. $\square$

The Lipschitz assumption is appropriate when the CATE function varies smoothly with the effect modifier. For threshold or step-function CATEs, L can be arbitrarily large and the bound becomes conservative; in such cases the clinical calibration should be interpreted as a worst-case bound rather than a tight estimate (see Section 5).

2.3 | Estimation

Given samples $\{X_{1,i}\}_{i=1}^{n_1}$ from region 1 and $\{X_{2,j}\}_{j=1}^{n_2}$ from region 2, nABCD is estimated using empirical distribution functions. Since $\hat{F}_1$ and $\hat{F}_2$ are piecewise constant (right-continuous step functions), the integral in equation (4) reduces to a finite sum over the intervals defined by the combined order statistics:

$$\widehat{\text{nABCD}} = \frac{\sum_{k=1}^{n_1+n_2-1} |\hat{F}_1(x_{(k)}) - \hat{F}_2(x_{(k)})| \cdot (x_{(k+1)} - x_{(k)})}{2 \cdot \widehat{\text{IQR}}_{\text{pooled}}}, \quad (8)$$

where $\hat{F}_r$ denotes the empirical CDF (right-continuous) of region r , and $x_{(1)} < \dots < x_{(n_1+n_2)}$ are the combined order statistics. The asymptotic distribution of W_1 is non-standard: del Barrio et al.¹¹ showed that $\sqrt{n} W_1(\hat{F}_n, F) \xrightarrow{d} \int |B(F(x))| dx$, where B is a standard Brownian bridge. Because the quantiles of this limiting distribution depend on the unknown F , no universal critical values are available and direct construction of asymptotic confidence intervals is impractical. We therefore employ the nonparametric percentile bootstrap with $B = 2,000$ replicates.¹² In the two-sample setting, when $F_1 \neq F_2$ the Hadamard derivative of the L_1 functional is linear, so the ordinary nonparametric bootstrap is consistent;¹³ convergence occurs at rate $\sqrt{n_1 n_2 / (n_1 + n_2)}$ (see Appendix A.2 for details). The estimator (8) is computed in $O((n_1 + n_2) \log(n_1 + n_2))$ time, dominated by sorting the combined order statistics.

2.4 | Interpretation and Clinical Calibration

A central feature of nABCD is its connection to treatment effect heterogeneity through Proposition 2. This connection provides the basis for a clinically grounded interpretation framework rather than reliance on fixed thresholds alone.

2.4.1 | Clinical Calibration via the Heterogeneity Bound

From equation (7), the maximum potential difference in regional average treatment effects attributable to effect modifier distributional differences is:

$$\Delta_{\max} = 2L \cdot \text{IQR}_{\text{pooled}} \cdot \text{nABCD}(F_1, F_2), \quad (9)$$

where L is the Lipschitz constant of the CATE function $\tau(x)$ with respect to the effect modifier. L serves as a sensitivity parameter controlling the worst-case treatment effect heterogeneity, and can be estimated from prior knowledge, pilot data, or published subgroup analyses.

The clinical calibration procedure is as follows:

1. Identify candidate effect modifiers and compute nABCD with bootstrap CIs for each effect modifier across region pairs.

2. For each effect modifier, estimate or bound L from prior knowledge (e.g., subgroup analyses showing treatment effect varies by $\Delta\tau$ per unit change in the effect modifier, giving $L \approx \Delta\tau/\Delta x$).
3. Compute $\Delta_{\max}$ using equation (9). This represents the worst-case treatment effect difference attributable to the distributional difference.
4. Derive the bootstrap CI for $\Delta_{\max}$ from the nABCD confidence interval: $[\Delta_{\max,L}, \Delta_{\max,U}] = 2L \cdot \text{IQR}_{\text{pooled}} \cdot [\text{nABCD}_L, \text{nABCD}_U]$. This expresses inferential uncertainty directly on the clinical scale.
5. Report $\Delta_{\max}$ and its CI alongside the clinical context—for example, the overall treatment effect size, the non-inferiority margin, or the minimal clinically important difference. This information enables trial designers and regulatory scientists to exercise clinical judgment about pooling, considering the totality of evidence rather than a binary accept/reject decision.

This estimation-centered approach deliberately avoids forcing pooling decisions into a binary hypothesis testing framework. The rationale is threefold. First, the ICH E17 guideline describes “similar enough” as inherently context-dependent, and reducing this judgment to a rejection threshold risks oversimplification. Second, the uncertainty in L means that $\Delta_{\max}$ itself is subject to sensitivity analysis (Section 4); a binary test cannot naturally accommodate this layered uncertainty. Third, consistent with recent calls to move beyond dichotomous statistical decisions,¹⁴ presenting $\Delta_{\max}$ with its CI provides richer information for regulatory deliberation than a p-value or a reject/fail-to-reject outcome.

When regulatory agencies require a formal decision rule, the estimation framework can be adapted: declare pooling acceptable if the upper bound of the 95% CI for $\Delta_{\max}$ falls below a pre-specified clinical margin Δ_{clin} . However, we recommend this as a supplementary rather than primary use of the nABCD framework.

The clinical calibration procedure is demonstrated with real clinical trial data in Section 4, where two candidate effect modifiers with contrasting levels of CATE sensitivity evidence illustrate both full calibration and planning-stage assessment workflows.

2.4.2 | Sensitivity Analysis When L Is Unknown

When L cannot be estimated from prior data—the most common situation at the MRCT planning stage—the clinical calibration framework supports two complementary approaches. First, the reference benchmarks (Table 1) provide an initial assessment of distributional magnitude without requiring L . Second, a reverse sensitivity analysis asks: *what value of L would be required for a given nABCD to produce a clinically meaningful $\Delta_{\max}$?* Rearranging equation (9):

$$L^* = \frac{\Delta_{\text{clin}}}{2 \cdot \text{IQR}_{\text{pooled}} \cdot \text{nABCD}}, \quad (10)$$

where Δ_{clin} is a pre-specified clinically important treatment effect difference (e.g., the overall treatment effect or the non-inferiority margin). If L^* exceeds the largest plausible CATE sensitivity for the effect modifier under consideration, the distributional difference is unlikely to produce meaningful treatment effect heterogeneity, regardless of the true L . This reverse calculation is demonstrated in Case A of Section 4.

2.4.3 | Reference Benchmarks

When prior knowledge of L is unavailable, the benchmarks in Table 1 provide a convenience reference for initial assessment. These labels describe *distributional magnitude* only and should not be conflated with clinical significance: as the calibration framework (equation 9) and Section 4 demonstrate, a “large” nABCD for a weak effect modifier may carry less clinical consequence than a “moderate” nABCD for a strong effect modifier, because CATE sensitivity L determines how distributional distance translates into treatment effect heterogeneity.

3 | SIMULATION STUDY

We conducted simulation studies to evaluate the estimation properties of the nABCD estimator: bias, variability, and coverage probability of bootstrap confidence intervals. These properties are essential for the clinical calibration framework, as reliable estimation of nABCD directly determines the quality of $\Delta_{\max}$ estimates. We assessed performance across a range of scenarios relevant to MRCT applications.

TABLE 1 Reference benchmarks for nABCD values when CATE sensitivity L is unknown.

nABCD Range	Distributional Magnitude	Calibration Guidance
< 0.05	Negligible	Clinical calibration unlikely to reveal meaningful $\Delta_{\max}$
0.05–0.15	Small	Calibration recommended if L is available
0.15–0.30	Moderate	Calibration important; interpret in clinical context
> 0.30	Large	Calibration essential before pooling deliberation

These benchmarks serve as initial reference points only; clinical interpretation should always proceed through the calibration framework of Section 2.4, as the same nABCD value may have different clinical implications depending on the Lipschitz constant L . Under Gaussian location shift, the benchmark boundaries correspond approximately to $\text{SMD} = 0.2, 0.5$, and 1.0σ (simulation scenarios S2, S3, and S4 in Table 2, with true nABCD values of 0.073, 0.180, and 0.328 respectively). Actual pooling decisions should use clinical calibration (equation 9) whenever possible.

3.1 | Simulation Design

3.1.1 | Scenarios

We designed eight systematic scenarios for methodological validation, examining controlled distributional differences across location, scale, shape, and combinations thereof. Table 2 summarizes these scenarios with their true nABCD values.

TABLE 2 Simulation scenarios with clinical motivations.

ID	Description	Distribution 1	Distribution 2	True nABCD
S1	Null (identical)	$N(50, 10^2)$	$N(50, 10^2)$	0.000
S2	Location 0.2σ	$N(50, 10^2)$	$N(52, 10^2)$	0.073
S3	Location 0.5σ	$N(50, 10^2)$	$N(55, 10^2)$	0.180
S4	Location 1.0σ	$N(50, 10^2)$	$N(60, 10^2)$	0.328
S5	Scale $1.5\times$	$N(50, 10^2)$	$N(50, 15^2)$	0.122
S6	Shape (Gamma)	$N(50, 10^2)$	$\text{Gamma}(25, 0.5)$	0.024
S7	Skew (Log-normal)	$N(50, 10^2)$	$\text{LogN}(\mu, \sigma)$	0.304
S8	Location + Scale	$N(50, 10^2)$	$N(55, 15^2)$	0.175

The Gamma distribution in S6 has shape parameter 25 and rate 0.5, yielding mean 50 and standard deviation 10. The Log-normal in S7 is parameterized to match mean 50 with $\text{CV} \approx 53\%$, typical of clinical biomarkers such as ALT. S8 combines location and scale differences, representing the most realistic MRCT pattern.

3.1.2 | Simulation Parameters

For each scenario, we generated samples of size $n = 50, 100$, and 200 per region, reflecting sample sizes typical in MRCT regional subgroups. We performed 10,000 replications per scenario–sample size combination to ensure stable estimates of operating characteristics, with Monte Carlo standard errors below 0.005 for all reported proportions. Bootstrap confidence intervals were computed using $B = 2,000$ resamples. All simulations were conducted in R version 4.5.1. True nABCD values for all scenarios were computed via Monte Carlo integration ($n_{\text{MC}} = 10^6$) using the mixture IQR, avoiding closed-form approximations that may use component rather than pooled IQR.

3.1.3 | Evaluation Metrics

Consistent with the estimation-centered framework, we evaluated:

1. *Bias*: $\text{Mean}(\widehat{\text{nABCD}}) - \text{nABCD}$
2. *Root mean squared error (RMSE)*: $\sqrt{\text{Mean}[(\widehat{\text{nABCD}} - \text{nABCD})^2]}$
3. *Coverage probability*: Proportion of 95% bootstrap CIs containing the true value
4. *CI width*: Mean width of bootstrap CIs, reflecting estimation precision

3.2 | Results

3.2.1 | Point Estimation and Coverage

Table 3 presents the bias of the nABCD estimator across scenarios and sample sizes. The estimator showed positive bias under the null hypothesis (S1), with bias decreasing from 0.093 at $n = 50$ to 0.047 at $n = 200$. This positive bias is attributable to the non-negative nature of the Wasserstein distance: even when true nABCD equals zero, sampling variability produces positive estimates.

TABLE 3 Bias of nABCD estimator by scenario and sample size.

Scenario	True nABCD	$n = 50$	$n = 100$	$n = 200$
S1 (Null)	0.000	0.093	0.066	0.047
S2 (0.2σ)	0.073	0.039	0.020	0.009
S3 (0.5σ)	0.180	0.010	0.003	0.001
S4 (1.0σ)	0.328	0.006	0.003	0.001
S5 (Scale)	0.122	0.028	0.014	0.007
S6 (Shape)	0.024	0.071	0.046	0.028
S7 (Skew)	0.304	0.015	0.008	0.003
S8 (Loc+Scale)	0.175	0.023	0.011	0.006

Bias is defined as $\text{Mean}(\widehat{\text{nABCD}}) - \text{nABCD}$. Results based on 10,000 replications with $B = 2,000$ bootstrap resamples.

For scenarios with true nABCD above approximately 0.1, bias was less than 0.02 in absolute value at $n \geq 100$, indicating satisfactory point estimation performance at practical sample sizes (Figure 2). For S4 (1.0σ location shift), bias was negligible (0.006 at $n = 50$, 0.001 at $n = 200$), confirming accurate estimation even for large distributional differences. For scenarios near the boundary (S1, S2, S6), positive bias was more pronounced and diminished slowly with sample size—a well-known phenomenon for non-negative statistics where sampling variability inflates estimates when the true value is close to zero. S6 (shape) exhibited the largest bias among non-null scenarios (0.071 at $n = 50$, 0.028 at $n = 200$), because its true nABCD (0.024) lies in the near-boundary regime where the non-negativity constraint dominates.

Table 4 presents coverage probabilities of the 95% bootstrap confidence intervals. For scenarios with true nABCD above approximately 0.1, coverage was near-nominal at $n \geq 100$: S3 (0.947–0.951), S4 (0.950–0.957), S7 (0.953–0.954), and S8 (0.935–0.944). S4 (1.0σ location shift) showed particularly stable coverage across all sample sizes, with no degradation at larger n , confirming satisfactory bootstrap performance even for large distributional differences. S5 (scale) showed moderate undercoverage at $n = 50$ (0.856) that improved to near-nominal at $n = 200$ (0.933). S2 (small location shift) showed undercoverage at $n = 50$ (0.652) due to its near-boundary true value, improving to 0.941 at $n = 200$.

Coverage for S6 (shape) is not reported because its true nABCD (0.024) is sufficiently small that positive bias dominates at all sample sizes considered. This is the same boundary phenomenon observed for S1 (null): when the true value lies near the parameter space boundary, the non-negativity constraint on nABCD produces persistent positive bias that prevents the confidence interval from containing the true value. This does not indicate estimator failure—the estimator is consistent—but rather that sample sizes well beyond 200 per region would be needed for reliable inference when the true distributional difference is negligible.

We also evaluated BCa (bias-corrected and accelerated) confidence intervals in preliminary experiments. BCa intervals showed consistently lower coverage than percentile intervals across all scenarios, particularly for near-boundary scenarios. The BCa correction overcorrects for nABCD because the statistic is bounded below by zero, causing the acceleration parameter to distort the quantile adjustment. Based on these findings, we adopt percentile bootstrap as the primary inference method.

Monte Carlo standard errors for all reported coverage probabilities were below 0.005 at 10,000 replications, ensuring that observed differences across scenarios and sample sizes reflect genuine operating characteristic differences rather than simulation noise.

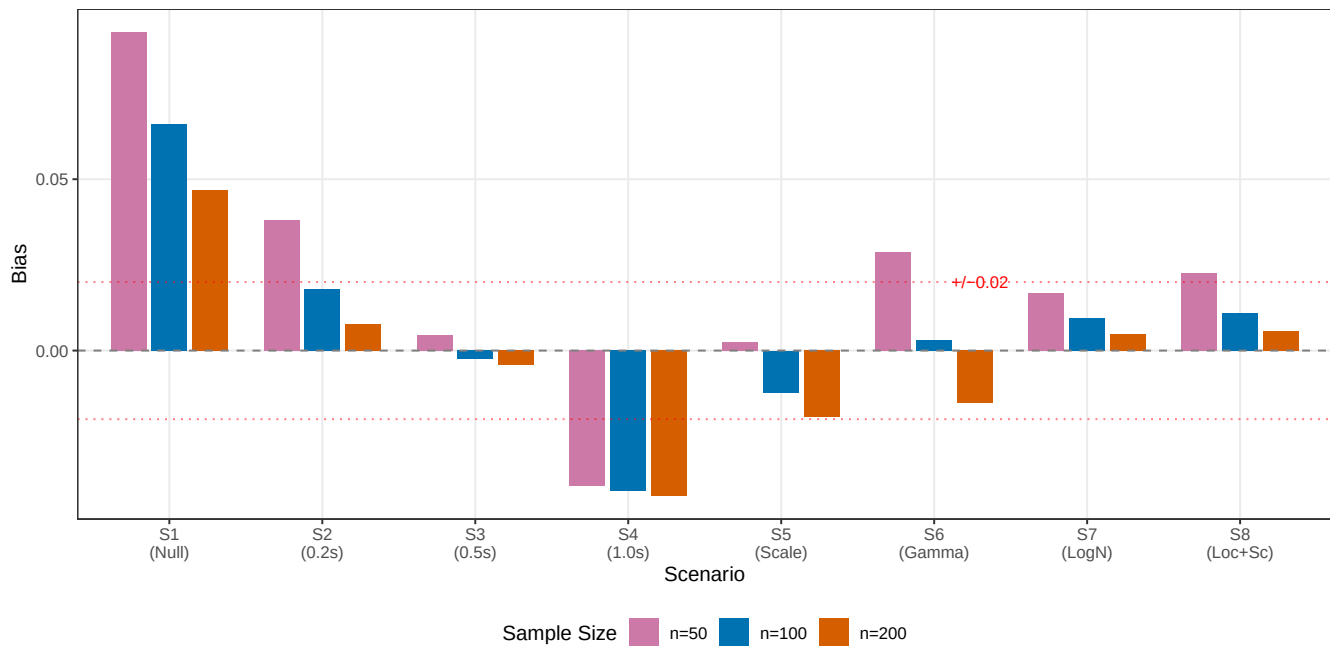


FIGURE 2 Bias of nABCD estimator by scenario and sample size. Horizontal dashed lines indicate ± 0.02 bias threshold. For scenarios with true nABCD above 0.1, bias is less than 0.02 at $n \geq 100$. Near-boundary scenarios (S1, S6) show larger positive bias due to the non-negativity constraint.

TABLE 4 Coverage probability of 95% percentile bootstrap confidence intervals.

Scenario	$n = 50$	$n = 100$	$n = 200$
S2 (0.2 σ)	0.652	0.881	0.941
S3 (0.5 σ)	0.947	0.951	0.947
S4 (1.0 σ)	0.950	0.950	0.957
S5 (Scale)	0.856	0.910	0.933
S7 (Skew)	0.953	0.954	0.954
S8 (Loc+Scale)	0.917	0.935	0.944

Coverage under S1 (Null) and S6 (Shape) is not reported because the true nABCD lies near the boundary of the parameter space (0 and 0.024, respectively), where positive bias from the non-negativity constraint prevents meaningful coverage assessment.

For S4, coverage was near-nominal across all sample sizes (0.950–0.957), confirming satisfactory performance even for large distributional differences. The undercoverage for S2 at small sample sizes (0.652 at $n = 50$) and S5 (0.856 at $n = 50$) reflects boundary proximity effects that diminish with increasing n .

3.2.2 | Estimation Precision

Figure 3 summarizes the estimation quality across scenarios and sample sizes, presenting both coverage probabilities and mean CI widths.

Table 5 presents the RMSE and mean CI width across scenarios. RMSE decreased consistently with increasing n , reaching values below 0.05 for all scenarios at $n = 200$ except S6 (shape), where the near-boundary bias inflated RMSE. Mean CI width narrowed proportionally, with CIs at $n = 100$ typically spanning 0.11–0.19 in nABCD units. In the clinical calibration framework, this CI width propagates to $\Delta_{\max}$: for example, with $L = 0.01/\text{pt}$ and $\text{IQR} = 11$ (values representative of NIHSS in the IST-3 application), a CI width of 0.13 in nABCD corresponds to a $\Delta_{\max}$ CI width of $2 \times 0.01 \times 11 \times 0.13 = 0.029$ (2.9%pt)—a level of precision sufficient for meaningful clinical calibration.

Under the null hypothesis (S1), the positive bias produces CIs that are shifted upward from zero. At $n = 50$, the mean estimate was 0.093 with mean CI width 0.16; at $n = 200$, the mean estimate decreased to 0.047 with mean CI width 0.08. This bias is relevant for clinical calibration: when the true distributional difference is zero, the estimator will suggest a small positive $\Delta_{\max}$. Practitioners should be aware of this conservative tendency, particularly at smaller sample sizes.

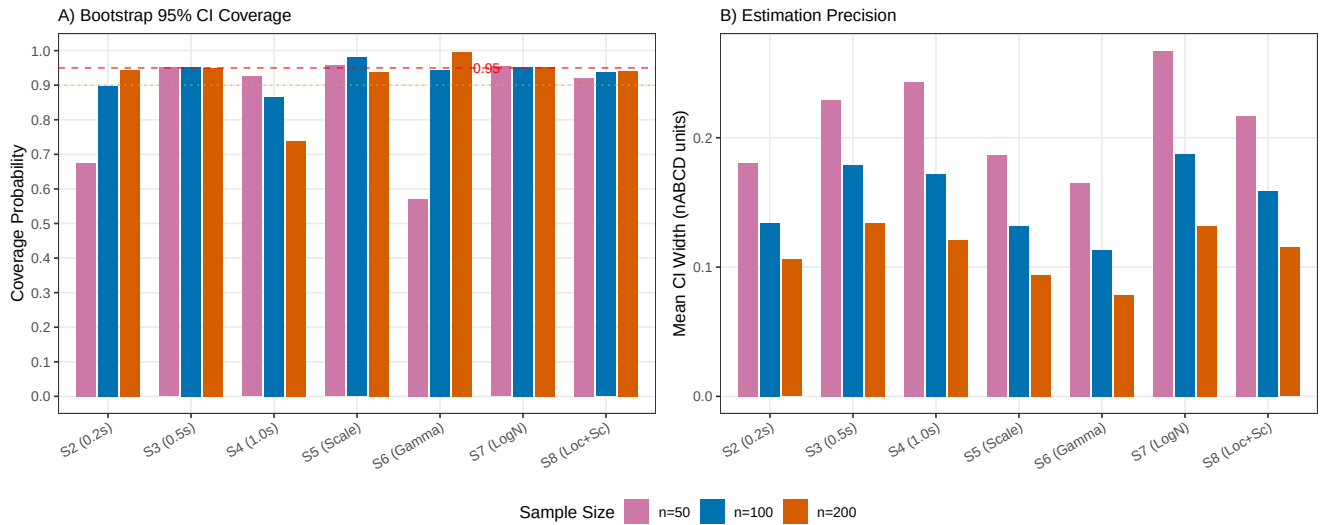


FIGURE 3 Estimation quality of nABCD bootstrap inference. Panel (A) shows coverage probability of 95% bootstrap confidence intervals; dashed and dotted lines indicate the nominal 0.95 and minimum acceptable 0.90 levels. Panel (B) shows mean CI width in nABCD units. Coverage exceeds 0.90 for most scenarios at $n \geq 100$; CI width narrows consistently with increasing n .

TABLE 5 RMSE and mean CI width by scenario and sample size.

Scenario	RMSE			Mean CI Width		
	$n = 50$	$n = 100$	$n = 200$	$n = 50$	$n = 100$	$n = 200$
S1 (Null)	0.099	0.069	0.049	0.16	0.11	0.08
S2 (0.2σ)	0.061	0.044	0.032	0.18	0.13	0.11
S3 (0.5σ)	0.067	0.048	0.035	0.23	0.18	0.13
S4 (1.0σ)	0.062	0.043	0.030	0.24	0.17	0.12
S5 (Scale)	0.053	0.036	0.025	0.19	0.13	0.09
S6 (Shape)	0.080	0.053	0.033	0.16	0.11	0.08
S7 (Skew)	0.070	0.049	0.034	0.27	0.19	0.13
S8 (Loc+Scale)	0.062	0.043	0.030	0.22	0.16	0.12

CI width is defined as mean of (upper – lower) across 10,000 replications.

3.2.3 | Comparison with Standardized Mean Difference

Table 6 compares the sensitivity of nABCD and SMD to different types of distributional differences. While both metrics respond to location shifts, SMD is insensitive to differences in variance and shape—the very types of distributional differences that may drive treatment effect heterogeneity through non-linear CATE functions.

TABLE 6 Sensitivity comparison: nABCD versus SMD at $n = 100$.

Scenario	nABCD (mean ± SD)	SMD (mean ± SD)	Implication
S3 (Location)	0.183 ± 0.048	0.50 ± 0.14	Both detect
S5 (Scale only)	0.136 ± 0.033	0.00 ± 0.14	Only nABCD detects
S6 (Shape only)	0.070 ± 0.025	0.00 ± 0.14	Only nABCD detects
S7 (Skew only)	0.312 ± 0.048	0.00 ± 0.14	Only nABCD detects

SMD depends only on the difference in means, making it insensitive to differences in variance, shape, and skewness. nABCD captures the full distributional difference. S7 is particularly informative: a large distributional difference (nABCD = 0.31) is entirely invisible to SMD.

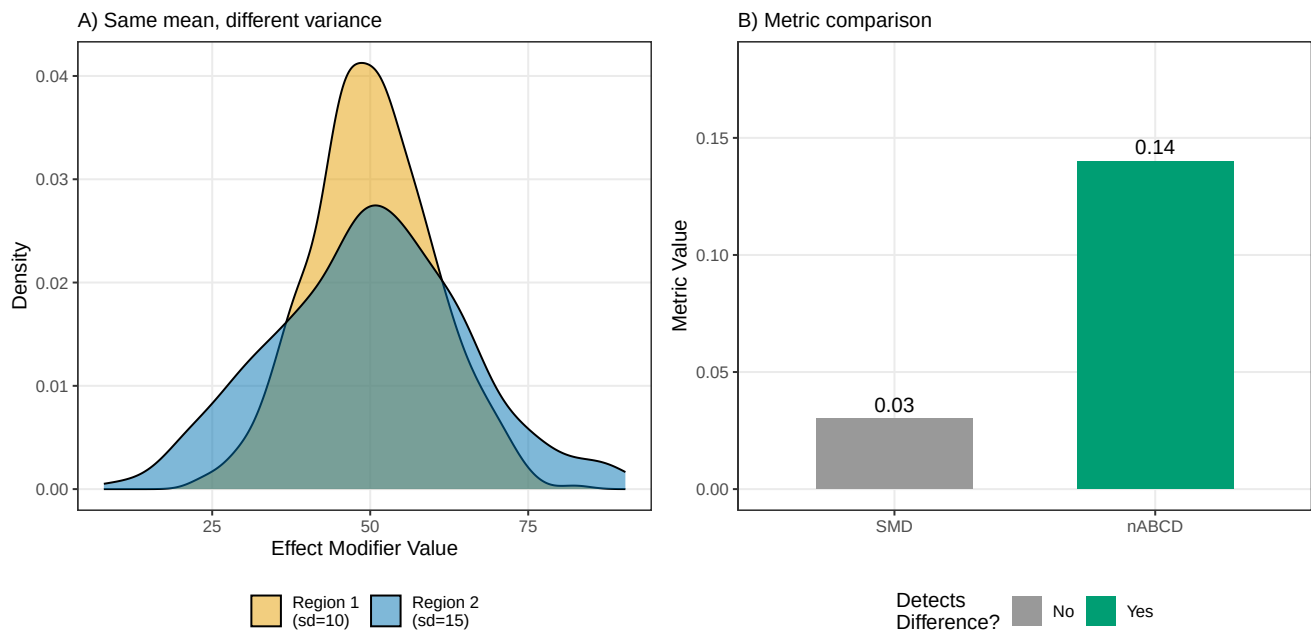


FIGURE 4 Comparison of nABCD and SMD across distributional difference types. nABCD responds to scale (S5) and shape (S6) differences where SMD remains near zero, demonstrating its advantage for full distributional comparison.

In summary, the simulation study across eight scenarios demonstrates that the nABCD estimator provides reliable estimation with the following properties at sample sizes typical in MRCT regional subgroups:

1. *Bias*: Less than 0.02 for most non-null scenarios at $n \geq 100$; positive bias is largest near the boundary (S1: 0.065, S6: 0.046 at $n = 100$) and negligible for scenarios with true nABCD above 0.1 (S3: 0.003, S4: 0.003 at $n = 100$)
2. *Coverage*: 0.88–0.96 at $n \geq 100$ for scenarios with true nABCD ≥ 0.1 (S3–S5, S7, S8); S7 (skew) and S8 (location + scale) achieved coverage between 0.93–0.95
3. *Precision*: RMSE below 0.06 and CI width below 0.19 at $n \geq 100$
4. *Sensitivity*: Captures location, scale, and shape differences where SMD is limited to location only

Scenario S3 (0.5 σ location shift, true nABCD = 0.180) exemplifies the operating characteristics at clinically relevant effect sizes: bias was negligible (+0.003 at $n = 100$), coverage was nominal (0.951), and CI width (0.18) translated to a $\Delta_{\max}$ CI width of $2 \times 0.01 \times 11 \times 0.18 = 0.040$ (4.0%pt) when calibrated with $L = 0.01/\text{pt}$ and IQR = 11 (NIHSS-scale parameters)—a level of precision sufficient for informed regulatory deliberation.

These estimation properties ensure that nABCD, when combined with the clinical calibration framework, provides $\Delta_{\max}$ estimates of sufficient quality for regulatory deliberation. We recommend $n \geq 100$ per region for reliable estimation and inference.

4 | APPLICATION: HYPOTHETICAL THROMBOLYTIC MRCT

We illustrate the nABCD framework through a hypothetical scenario grounded in publicly available data. A sponsor is developing a novel thrombolytic agent (drug A) for acute ischaemic stroke and is planning a Phase 3 MRCT. At this planning stage, the sponsor requires regional effect modifier distribution data to inform pooling decisions. Such data need not come from clinical trials: disease registries (e.g., SITS-MOST for stroke), electronic health records, epidemiological databases, or any representative population data source would suffice. Here, we use publicly available individual patient data (IPD) from two international stroke trials as convenient, well-documented data sources. Based on the Emberson et al.¹⁵ IPD meta-analysis of nine alteplase trials ($n = 6,756$), NIHSS baseline stroke severity (treatment $\times$ effect modifier interaction $p = 0.06$) and age ($p = 0.53$) are identified

as candidate effect modifiers for the thrombolytic class. We emphasize that nABCD serves as one piece of evidence supporting pooling decisions, not a standalone decision criterion.

4.1 | Application Data and Scenario Design

To illustrate both common planning scenarios, we employ two complementary historical datasets from the same stroke trial program (conducted at the University of Edinburgh). A sponsor developing a thrombolytic agent for acute ischaemic stroke faces the question: “Which regional distributions of effect modifiers would warrant modifying our pooling strategy?” The answer depends on what is known about CATE sensitivity:

- **Case A (Effect Modifier Unknown):** When prior evidence on treatment $\times$ effect modifier interactions is absent or non-significant, distributional heterogeneity alone must guide decisions. This is the most common situation in MRCT planning.
- **Case B (Effect Modifier Identified):** When prior evidence confirms effect modifier status with estimable CATE sensitivity, clinical calibration becomes possible, translating distributional differences into clinical impact.

IST-1 (Case A): The First International Stroke Trial (1991–1996) randomized 19,435 patients across 31 countries to aspirin, heparin, or control.¹⁶ The trial was not designed to detect effect modification; consequently, Chen et al. found no statistically significant treatment $\times$ effect modifier interactions across 28 subgroups.¹⁷ IST-1’s global scope enables assessment of geographic distributional heterogeneity with minimal prior assumptions about CATE sensitivity.

IST-3 (Case B): The Third International Stroke Trial (2000–2012) randomized 3,035 patients with acute ischaemic stroke to alteplase ($n = 1,515$) or control ($n = 1,520$) across 12 countries.¹⁸ The Emberson et al.¹⁵ IPD meta-analysis identified confirmed effect modifiers in thrombolytic trials: NIHSS (interaction $p = 0.06$), age ($p = 0.53$), and treatment delay ($p = 0.016$). IST-3 provides the necessary prior evidence to estimate CATE sensitivity and calibrate nABCD into clinically meaningful $\Delta_{\max}$ values. We included the eight IST-3 countries with $n \geq 50$ patients (Table 7). Belgium ($n = 73$, 2.4% of total enrolment) serves as the anchor country in this demonstration—its small sample size parallels the situation that small-sample regions such as Japan or China may face in global MRCTs. The exercise of identifying Belgium’s suitable pooling partners simulates the decision a sponsor would face when seeking regulatory-compliant regional efficacy assessment through pooling.

Both trials serve as methodological illustrations of the nABCD framework. Neither was designed as an MRCT, but their publicly available IPD provides representative baseline effect modifier distributions across diverse geographic regions—precisely the type of data a sponsor would assemble, from any available source, during MRCT planning. Together they exemplify the decision environment sponsors face during actual Phase 3 planning.

TABLE 7 IST-3 baseline characteristics by country.

Country	n	Age, mean (SD)	Delay, mean (SD)	NIHSS, mean (SD)
UK	1,447	78.0 (12.2)	4.1 (1.2)	13.1 (6.9)
Poland	347	73.7 (13.1)	4.6 (1.2)	9.6 (6.4)
Italy	326	75.9 (12.2)	4.2 (1.1)	11.1 (6.6)
Sweden	297	81.0 (10.8)	3.8 (1.3)	10.0 (6.5)
Norway	204	76.1 (10.3)	4.3 (1.8)	11.9 (6.9)
Australia	179	74.7 (13.1)	4.6 (1.1)	14.4 (7.5)
Portugal	82	79.0 (11.5)	4.3 (1.2)	14.8 (6.0)
Belgium	73	76.1 (12.4)	4.0 (1.1)	10.9 (6.1)

UK contributed 48% of the total sample. Belgium ($n = 73$) and Portugal ($n = 82$) fall below the recommended minimum of $n \geq 100$ per region; bootstrap CIs for pairs involving these countries should be interpreted with caution.

4.2 | Case A: Effect Modifier Unknown (IST-1)

In the most common planning scenario, sponsors lack confirmed prior evidence on CATE sensitivity L . The First International Stroke Trial (IST-1; 1991–1996) exemplifies this setting: despite involving 19,435 patients across 31 countries, no statistically significant treatment $\times$ age, treatment $\times$ SBP, or treatment $\times$ consciousness level interactions were detected (all $p > 0.05$ across 28 subgroups),¹⁷ meaning L is not estimable from the primary efficacy endpoint. This absence of confirmed effect modifier evidence is realistic: Song et al. noted that “It is extremely challenging to identify true effect modifiers” in published trial data,⁵ suggesting that most MRCT planning occurs in the effect-modifier-unknown setting.

For Case A, nABCD’s role is purely distributional: to quantify geographic heterogeneity in effect modifier distributions and assess whether that heterogeneity might plausibly translate to treatment effect heterogeneity through non-linear mechanisms. Clinical calibration is replaced by L^* sensitivity analysis: what value of CATE sensitivity would make the observed distributional difference clinically concerning?

IST-1 provides global scope (31 countries across five continents including India, Singapore, and Hong Kong), enabling assessment of distributional patterns across diverse healthcare contexts. We computed nABCD for all 465 country pairs across three candidate effect modifiers: age, regional systolic BP (RSBP), and treatment delay.

4.2.1 | Geographic Distributional Heterogeneity

The breadth of IST-1’s geographic coverage reveals substantial regional heterogeneity in effect modifier distributions. For age, the India–UK pair exhibited maximum nABCD of 0.565 (approximately fourfold larger than the maximum observed in IST-3), driven by demographic differences in stroke population composition across healthcare systems. Asian countries showed progressively smaller age-related distributional differences: India (mean nABCD = 0.375), Singapore (0.205), and Hong Kong (0.121), reflecting both demographic variation and regional healthcare access patterns. This global-scale variation in effect modifier distributions exemplifies a key motivation for Case A: when effect modifiers cannot be studied for interaction, distributional heterogeneity provides the primary evidence base for pooling decisions. The 31-country coverage enables assessment of whether observed nABCD values reflect meaningful geographic clustering (e.g., Asia vs. Europe) or random variation.

For RSBP, the Pearson correlation between ISMDI and nABCD across 465 pairs was 0.906—the lowest observed in IST-1, indicating greater scale and shape heterogeneity than for age. For treatment delay, both central tendency and distributional shape varied markedly: Ireland exhibited exceptionally long mean delays (approximately 33 hours), while Scandinavian countries showed shorter mean delays with high variability (kurtosis > 10 in some regions), indicating that simple mean-based comparisons obscure the underlying distributional asymmetry that nABCD captures.

4.2.2 | Sensitivity Analysis: L^* Thresholds

For maximum-nABCD pairs identified in IST-1, we conducted L^* reverse-calculation analysis to determine what CATE sensitivity value would translate the observed distributional difference into clinically meaningful heterogeneity. For the India–UK age comparison (nABCD = 0.565), the L^* threshold for $\Delta_{\max} = 2\%$ pt was approximately 0.005/year—a modest value suggesting that even small age-related treatment interactions could plausibly result in clinically concerning regional heterogeneity if they exist. This approach operationalizes the principle that “clinical meaningfulness depends on context”: rather than applying fixed nABCD thresholds, sponsors can evaluate whether plausible CATE sensitivities would render the observed distributional difference clinically concerning. When L^* is implausibly large, the distributional difference has low clinical risk; when L^* is modest (as in the India–UK case), the geographic heterogeneity warrants explicit mitigation strategies in trial design.

4.3 | Case B: Effect Modifier Identified (IST-3)

When prior evidence confirms effect modifier status—as is the case for treatment timing and early NIHSS in thrombolytic trials—clinical calibration becomes possible. The Third International Stroke Trial (IST-3; 2000–2012) provides 3,035 patients across 8 countries with multiple candidate effect modifiers where interaction p -values have been published. This scenario allows direct translation of distributional differences (nABCD) into clinical impact ($\Delta_{\max}$) through the heterogeneity bound.

4.3.1 | Distributional Comparison: nABCD versus SMD

We computed nABCD and SMD⁷ for all 28 country pairs across NIHSS, age, and treatment delay. Table 8 summarizes the nABCD results.

TABLE 8 nABCD summary across 28 pairwise country comparisons in IST-3.

Candidate effect modifier	Skewness	Min	Median	Mean	Max	Max pair
Age (years)	−1.26	0.039	0.103	0.123	0.285	Sweden–Belgium
NIHSS (score)	0.49	0.027	0.101	0.113	0.240	Poland–Portugal
Treatment delay (h)	1.21	0.026	0.098	0.107	0.195	Sweden–Australia

Skewness is computed over the pooled sample. Treatment delay exhibits the highest positive skewness and excess kurtosis ($= 20.0$), with extreme heterogeneity across countries (Norway: skewness = 6.76, SD = 1.80 vs. Italy: skewness = -0.05 , SD = 1.09).

Treatment delay provides the most informative comparison of nABCD and SMD, because its highly skewed, heavy-tailed distribution creates situations where location-based metrics are misleading. The Pearson correlation between |SMD| and nABCD across the 28 pairs was 0.91 for treatment delay, compared with 0.95 for age and 0.98 for NIHSS—the lower correlation reflecting treatment delay’s greater scale and shape heterogeneity across countries.

The Norway–Portugal pair is particularly illustrative: the two countries had virtually identical mean treatment delays (4.34 vs. 4.33 hours; SMD = 0.007), yet nABCD was 0.069—indicating a non-trivial distributional difference invisible to SMD. This divergence arises because Norway’s treatment delay distribution is dominated by an extreme right tail (skewness = 6.76, range up to 24.3 hours, SD = 1.80), while Portugal’s distribution is more compact (skewness = -0.23 , SD = 1.21). SMD, which depends only on the difference in means, treats these distributions as essentially identical. The Poland–Norway pair further illustrates the pattern: despite a moderate SMD of 0.151, nABCD reached 0.115 (ratio 0.76), reflecting the large scale difference (SD = 1.16 vs. 1.80) between the two countries. These cases parallel the simulation findings from scenarios S5 (scale) and S7 (skew), where nABCD detected distributional differences that SMD missed entirely.

For age and NIHSS, nABCD and |SMD| were more strongly correlated, because country-level differences were driven primarily by location (mean) shifts with relatively homogeneous variance. This is expected: when distributions are approximately normal with equal variances, nABCD and SMD convey similar information. The practical advantage of nABCD emerges precisely when an effect modifier’s distribution departs from normality—as is common for biomarkers, time-to-event variables, and clinical scores with floor or ceiling effects.

4.3.2 | Clinical Calibration: NIHSS

NIHSS has the strongest prior evidence as an effect modifier: Emberson et al.¹⁵ reported a treatment $\times$ NIHSS interaction $p = 0.06$, and the IST-3 data show $p = 0.001$, indicating that CATE sensitivity L is estimable with reasonable precision. For drug A, the true L is unknown. We borrow the alteplase-based estimate from IST-3 as a same-class reference and conduct sensitivity analysis. Using IST-3’s logistic regression with treatment $\times$ NIHSS interaction (outcome: OHS 0–2 at 6 months), followed by marginal standardization, we estimated $L_{\text{mean}} = 0.00950/\text{pt}$ and $L_{\text{max}} = 0.01398/\text{pt}$ (Table 9).

TABLE 9 Clinical calibration of nABCD: CATE sensitivity estimation and Δ_{max} .

Candidate effect modifier	Interaction p	L_{max}	L_{mean}	nABCD _{max}	$\Delta_{\text{max}} (L_{\text{max}})$	$\Delta_{\text{max}} (L_{\text{mean}})$
NIHSS	0.001	0.01398/pt	0.00950/pt	0.240	7.37%pt	5.02%pt
Age	0.614	0.00090/yr	0.00065/yr	0.285	0.65%pt	0.47%pt

Δ_{max} expressed in risk difference percentage points. Overall treatment effect RD $\approx +1.5\%$ pt. L_{max} : maximum gradient over effect modifier grid; L_{mean} : average gradient. Δ_{max} computed for the maximum nABCD across all 28 country pairs for each effect modifier. Interaction p -values from IST-3 data.

For the maximum-nABCD pair (Poland–Portugal, nABCD = 0.240), Δ_{max} under the alteplase-based L_{mean} is 5.02%pt—substantially exceeding the overall treatment effect (RD $\approx 1.5\%$ pt). To assess robustness to the class-transferability assumption,

we varied L from $0.5\times$ to $2\times$ the alteplase estimate: at $0.5L_{\text{mean}}$, $\Delta_{\text{max}} = 2.51\% \text{pt}$; at $2L_{\text{mean}}$, $\Delta_{\text{max}} = 10.04\% \text{pt}$. Across this range, NIHSS distributional differences translate to clinically meaningful potential treatment effect heterogeneity, warranting attention in the pooling strategy regardless of the precise L value for drug A.

4.3.3 | Distributional Assessment: Age

Age has weak evidence as an effect modifier: Emberson et al.¹⁵ reported $p = 0.53$, and the IST-3 data confirm non-significance ($p = 0.614$). The estimated CATE sensitivity is small ($L_{\text{mean}} = 0.00065/\text{yr}$) with wide uncertainty. Because L is not statistically distinguishable from zero, the following calibration is illustrative—demonstrating how the framework behaves for a weak effect modifier—rather than a basis for clinical decision-making.

Given the weak effect modifier evidence, the primary assessment uses nABCD with the reference benchmarks (Table 1). The maximum pairwise nABCD for age was 0.285 (Sweden–Belgium), falling in the “moderate” range. Despite exhibiting the largest distributional difference of any candidate effect modifier, the clinical calibration yields a small Δ_{max} : 0.47%pt using L_{mean} , or 0.65%pt using L_{max} (Table 9). Even at $2L_{\text{mean}}$, Δ_{max} remains 0.94%pt—below the overall treatment effect. Age distributional differences across the planned regions are unlikely to generate clinically meaningful treatment effect heterogeneity.

4.4 | Practical Implications and Framework Integration

The two scenarios above illustrate the flexibility and practical scope of the nABCD framework across the most common planning situations:

Case A (Effect Modifier Unknown): When Prior Evidence is Unavailable

When L cannot be estimated (as in IST-1 aspirin data), nABCD with L^* sensitivity analysis provides the primary decision tool. The sponsor computes nABCD across all candidate effect modifiers and region pairs, then conducts sensitivity analysis to determine “What CATE sensitivity value would make this distributional difference clinically meaningful?” If the required L^* is implausibly large, the distributional difference can be regarded as unlikely to generate clinically meaningful heterogeneity. If L^* is modest, the distributional heterogeneity identified by nABCD warrants explicit mitigation (e.g., stratified randomization, protocol restrictions, subgroup monitoring).

Case B (Effect Modifier Identified): When Prior Evidence is Available

When L is estimable from prior trials (as in IST-3 NIHSS with $p = 0.001$), clinical calibration translates nABCD directly into Δ_{max} on the clinical outcome scale. The contrast between NIHSS and age exemplifies a central principle: the same nABCD magnitude can have vastly different clinical consequences depending on CATE sensitivity. Age had the largest distributional difference (nABCD = 0.285) yet the smallest clinical impact ($\Delta_{\text{max}} = 0.47\% \text{pt}$), while NIHSS had a moderate distributional difference (nABCD = 0.240) but an order-of-magnitude larger clinical impact ($\Delta_{\text{max}} = 5.02\% \text{pt}$). This underscores that nABCD magnitude alone is insufficient for pooling decisions; clinical calibration provides the necessary context.

The Distributional Advantage of nABCD Across Both Scenarios

Treatment delay exemplifies nABCD’s distributional sensitivity most clearly. Treatment delay is a confirmed effect modifier from the Emberson et al. IPD meta-analysis of nine alteplase trials ($p = 0.016$); however, IST-3 alone lacks power to detect this interaction ($p = 0.567$ within a single trial of $n = 3,035$). This discrepancy between the meta-analytic and single-trial evidence is typical of candidate effect modifiers at the planning stage: the sponsor’s prior belief in effect modifier status may rest on external evidence that the planning dataset cannot independently confirm. Despite this uncertainty, treatment delay’s highly skewed distribution renders SMD unreliable for assessing cross-regional similarity. In IST-3, the Norway–Portugal pair had $\text{SMD} \approx 0$ yet nABCD = 0.069, a divergence driven entirely by scale and shape differences. A sponsor relying solely on SMD would conclude that treatment delay distributions are identical across these regions, potentially overlooking a source of treatment effect

heterogeneity. The distributional asymmetry detected by nABCD warrants explicit acknowledgment in the pooling rationale regardless of the interaction p -value, because nABCD's value lies in detecting distributional features that *could* translate to heterogeneity if the effect modifier relationship holds.

Several limitations of this application should be noted:

1. *Illustrative data sources*: IST-1 and IST-3 were not designed as MRCTs; they serve as convenient publicly available data sources for methodological demonstration. In practice, sponsors would assemble regional effect modifier distributions from the most representative and current sources available, which may include disease registries, electronic health records, or other RWE databases.
2. *Small-sample countries*: Belgium ($n = 73$) and Portugal ($n = 82$) fall below the recommended $n \geq 100$; pairs involving these countries have wider bootstrap CIs.
3. *Transferability of L* : CATE sensitivity was estimated from alteplase data, not drug A; the class-transferability assumption requires justification and sensitivity analysis as demonstrated above.
4. *Temporal generalizability*: IST data span 1991–2011; demographic shifts in stroke populations may limit applicability to current trial planning. This limitation is specific to the illustrative datasets, not to the nABCD framework itself, which can be applied to contemporary data sources.
5. *Geographic coverage*: Countries in IST-1/IST-3 may not match planned Phase 3 regions; the effect modifier distributions serve only as historical reference.

5 | DISCUSSION

We developed nABCD, a normalized dissimilarity index that combines Wasserstein-1 distance with IQR normalization to capture full distributional differences—including scale, shape, and skewness—that conventional measures such as SMD miss. The clinical calibration framework translates these distributional differences into context-specific $\Delta_{\max}$ assessments when CATE sensitivity L is estimable. The IST-3 application demonstrated a key practical implication: rankings by distributional magnitude and clinical impact can diverge—age exhibited the largest nABCD (0.285) yet smallest $\Delta_{\max}$ (0.47%pt), while NIHSS showed moderate nABCD (0.240) but substantially larger $\Delta_{\max}$ (5.02%pt)—underscoring that clinical calibration, not nABCD magnitude alone, should guide pooling decisions.

Compared to existing approaches, nABCD offers three advantages. First, it captures distributional differences invisible to SMD: our simulation confirmed that scale (S5), shape (S6), and skewness (S7) differences produced near-zero SMD but non-negligible nABCD, and the IST-3 application revealed the same pattern with real data (treatment delay: SMD ≈ 0 between Norway and Portugal, nABCD = 0.069). Second, unlike the KS statistic, nABCD connects directly to treatment effect heterogeneity through the heterogeneity bound (equation 9), enabling clinical calibration rather than purely statistical assessment. Third, unlike visual inspection, nABCD is objective, reproducible, and amenable to bootstrap inference. Our approach is conceptually distinct from treatment effect consistency assessment methods² which evaluate whether observed treatment effects agree across regions; nABCD operates upstream, quantifying similarity in the effect modifier distributions that may drive such heterogeneity.

The clinical calibration procedure introduced in Section 2.4 represents a departure from fixed-threshold approaches. Cohen's d benchmarks (small = 0.2, medium = 0.5, large = 0.8) have been widely criticized for context-free application, and we deliberately avoid this pattern. Instead, the heterogeneity bound provides a principled mechanism to translate nABCD into clinically meaningful units, respecting the ICH E17 principle that “similar enough” is context-dependent. A natural consequence of clinical calibration is that rankings by distributional distance and clinical impact may differ—as observed in IST-3, where age had the largest nABCD yet the smallest $\Delta_{\max}$ —because the same distributional difference carries different clinical weight depending on CATE sensitivity. More broadly, the IST-3 application demonstrated that the same nABCD framework serves different purposes depending on the availability of CATE sensitivity evidence: for NIHSS, where L was estimable, full calibration translated distributional differences into clinically interpretable $\Delta_{\max}$ values; for age, where L was uncertain, nABCD with reference benchmarks (Table 1) provided the primary assessment of distributional similarity. This connection draws on transportability theory^{19,20} to relate distributional distance to potential treatment effect heterogeneity.

Based on our findings, we offer the following recommendations for practitioners:

1. Compute nABCD with bootstrap CIs ($n \geq 100$ per region) for each candidate effect modifier across region pairs.

2. Translate nABCD into $\Delta_{\max}$ and its CI on the clinical outcome scale using equation (9), incorporating prior knowledge or estimates of CATE sensitivity L . When L is estimated from data, report both $L_{\max}$ (conservative upper bound) and L_{mean} (realistic estimate), as in the IST-3 application (Section 4).
3. Report $\Delta_{\max}$ and its CI alongside the overall treatment effect, non-inferiority margin, or other clinically relevant benchmarks to support informed deliberation about pooling.
4. Conduct sensitivity analyses over plausible values of L when prior knowledge is uncertain.
5. When L cannot be estimated, use the reference benchmarks in Table 1 as initial guidance, recognizing that clinical calibration provides a more rigorous basis for decision-making.

In practice, multiple candidate effect modifiers may be evaluated simultaneously. The nABCD framework assesses each effect modifier separately, and we recommend reporting $\Delta_{\max}$ and its CI for every candidate effect modifier across all region pairs. To arrive at an overall pooling recommendation, trial designers may adopt a conservative approach based on the maximum $\Delta_{\max}$ across all effect modifiers and region pairs, or they may take a totality-of-evidence approach in which the collection of $\Delta_{\max}$ values, together with their uncertainties and the reliability of each L estimate, informs a holistic judgment. The choice between these strategies depends on the regulatory context: for applications where a formal pooling criterion is required, the maximum $\Delta_{\max}$ provides a defensible worst-case assessment; for advisory settings where pooling decisions are informed by clinical deliberation, the full set of calibrated results offers a richer evidence base.

A distinctive practical advantage of the nABCD framework is its compatibility with diverse data sources. Because nABCD requires only baseline effect modifier distributions—not treatment outcomes—regional distributional data can be assembled from prior clinical trials, disease registries, electronic health records, epidemiological databases, or other real-world evidence (RWE) sources. This data-source flexibility is increasingly relevant as regulatory agencies promote the use of RWE in drug development (ICH E6(R3)). The representativeness of the data source with respect to the target trial population is a prerequisite for valid nABCD estimation, regardless of the data origin.

Application to Japanese and Chinese regulatory contexts. The nABCD framework is particularly relevant for regulatory submissions requiring regional subpopulation consistency.^{4,5} In global MRCTs, regional enrolment is often limited, making direct demonstration of consistency difficult. Matsushima et al.'s⁴ PMDA workshop documented that regional imbalance in CRP-positive/MRI-negative status caused apparent inconsistency in the secukinumab MRCT, underscoring the practical consequences when effect modifier distributional differences are not assessed at the planning stage. nABCD provides the quantitative tool to identify such distributional differences prospectively, enabling sponsors to select pooling partners that minimize the risk of apparent inconsistency. As demonstrated in the IST-3 application using Belgium ($n = 73$) as a small-sample anchor, the framework directly simulates the decision environment faced by sponsors seeking regulatory-compliant regional assessment through pooling.

The practical availability of CATE sensitivity L varies across settings. When L is estimable from prior trials or meta-analyses in the same therapeutic area, full clinical calibration is possible, as demonstrated for NIHSS in the IST-3 application. At the MRCT planning stage, however, L is often uncertain—particularly for novel agents or therapeutic areas lacking subgroup analyses. In such cases, nABCD with reference benchmarks (Table 1) provides the primary assessment of distributional similarity, supplemented by sensitivity analysis over plausible L ranges. Importantly, nABCD's core value as a distributional measure—capturing scale, shape, and skewness differences invisible to SMD—holds regardless of whether L is available; clinical calibration enhances but does not replace this distributional assessment.

Several limitations should be acknowledged:

1. The current formulation applies to continuous effect modifiers only; extensions to categorical or mixed-type effect modifiers require further development.
2. nABCD evaluates each effect modifier separately rather than jointly; a multivariate extension would address potential confounding among effect modifiers.
3. Positive bias under the null and near-boundary scenarios (true nABCD $\lesssim 0.05$) can inflate nABCD estimates and prevent nominal coverage. This is inherent to the non-negative nature of nABCD as a function of the Wasserstein distance: when the true value lies at or near the boundary of the parameter space, sampling variability produces strictly positive estimates even when the true distributional difference is negligible.²¹ In our simulation, S6 (shape; true nABCD = 0.024) exhibited substantial positive bias (0.071 at $n = 50$, 0.028 at $n = 200$) and zero coverage at all sample sizes. We therefore do not report coverage for near-boundary scenarios and recommend interpreting nABCD estimates near zero with caution. For scenarios with true nABCD above approximately 0.1, the parameter lies in the interior of the parameter space where standard bootstrap

consistency holds, and our simulation confirms this with coverage of 0.88–0.96 at $n \geq 100$ across S2–S5, S7, and S8; S4 (1.0 σ location shift; true nABCD = 0.328) achieved near-nominal coverage (0.950–0.957), confirming reliable inference even for large distributional differences. We use the percentile bootstrap, which is first-order accurate; bias-corrected methods such as BCa showed inferior performance for this bounded statistic (Section 3.2), and the studentized bootstrap would require variance estimation for the Wasserstein distance ratio, which adds substantial complexity. We recommend $n \geq 100$ per region for reliable point estimation and confidence intervals.

4. The clinical calibration requires estimation of CATE sensitivity L , which may not always be available from prior data. Reference benchmarks are provided as a fallback but should not replace context-specific calibration.
5. When multiple effect modifiers are relevant, the current framework evaluates each separately. Aggregation strategies for combining nABCD values across effect modifiers—such as weighted combinations reflecting clinical importance or multivariate extensions using higher-dimensional optimal transport—remain topics for future investigation.
6. As with any covariate-based approach, nABCD assesses similarity only with respect to measured effect modifiers. Unmeasured or unknown effect modifiers may contribute to treatment effect heterogeneity beyond what the index captures.
7. The clinical calibration demonstrated in the IST-3 application used CATE sensitivity estimates from alteplase; transferability of L to other agents in the same pharmacological class is an assumption that requires clinical judgment and may benefit from sensitivity analysis.
8. The IST-3 data were collected during 2000–2011; demographic and clinical practice changes may limit the applicability of these country-level effect modifier distributions to contemporary trial planning.

Future work should address multivariate extensions, bias correction methods for small samples, and empirical calibration of CATE sensitivity parameters using real MRCT data. Methods for estimating L from historical trial databases would strengthen the clinical calibration framework. The nABCD framework could also be extended to longitudinal effect modifier profiles, enabling dynamic assessment of distributional similarity over time.

An R package implementing the nABCD index with bootstrap inference and the clinical calibration framework described herein is under development.

nABCD fills a methodological gap in ICH E17 implementation by providing a principled framework for assessing distributional similarity. Rather than reducing “similar enough” to a fixed threshold, the clinical calibration procedure translates distributional differences into context-specific assessments of potential treatment effect heterogeneity, enabling evidence-based and clinically grounded pooling decisions. Open-source R code is available to facilitate adoption.

AUTHOR CONTRIBUTIONS

[Author 1] conceived the research question and led the project. [Author 2] developed the mathematical framework and conducted the simulation study. [Author 3] contributed to the application example and manuscript preparation. All authors reviewed and approved the final manuscript.

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FINANCIAL DISCLOSURE

None reported.

CONFLICT OF INTEREST

The authors declare no potential conflict of interests.

DATA AVAILABILITY STATEMENT

R code for computing nABCD and reproducing the simulation study is available at [repository URL]. The IST-3 individual patient data are publicly available from Edinburgh DataShare.¹⁸

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SUPPORTING INFORMATION

Additional supporting information may be found in the online version of the article at the publisher’s website. Supporting materials include: complete R code for nABCD computation, simulation scripts, and additional figures for realistic clinical scenarios.

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APPENDIX

A PROOFS AND MATHEMATICAL DETAILS

A.1 Proof of Proposition 1

Non-negativity follows directly from the non-negativity of the Wasserstein-1 distance: $W_1(F_1, F_2) = \int |F_1(x) - F_2(x)| dx \geq 0$. Since $\text{IQR}_{\text{pooled}} > 0$ for non-degenerate distributions, the ratio $\text{nABCD} = W_1/(2 \cdot \text{IQR}_{\text{pooled}}) \geq 0$.

For the identity of indiscernibles, we show both directions. If $F_1 = F_2$, then $|F_1(x) - F_2(x)| = 0$ for all x , so $W_1(F_1, F_2) = 0$ and hence $\text{nABCD} = 0$. Conversely, if $\text{nABCD} = 0$, then $W_1(F_1, F_2) = \int |F_1(x) - F_2(x)| dx = 0$. Since $|F_1(x) - F_2(x)| \geq 0$ and is right-continuous, $\int |F_1(x) - F_2(x)| dx = 0$ implies $F_1(x) = F_2(x)$ almost everywhere; by right-continuity of CDFs, $F_1 = F_2$ everywhere.

A.2 Asymptotic Properties

Under standard regularity conditions, the empirical Wasserstein-1 distance $W_1(\hat{F}_1, \hat{F}_2)$ is a consistent estimator of $W_1(F_1, F_2)$.¹¹ Combined with the consistency of the empirical IQR, $\widehat{\text{nABCD}}$ is a consistent estimator of $\text{nABCD}(F_1, F_2)$.

In one dimension, the empirical W_1 distance converges at rate $O(n^{-1/2})$ without logarithmic correction.¹¹ For two-sample estimation with $F_1 \neq F_2$ and finite second moments, $\sqrt{n}(W_1(\hat{F}_{1,n_1}, \hat{F}_{2,n_2}) - W_1(F_1, F_2))$ converges in distribution to a mean-zero

Gaussian, where $n = n_1 + n_2$ with $n_1/n \rightarrow \lambda \in (0, 1)$. Since the empirical IQR also converges at rate $O(n^{-1/2})$ under standard density conditions, the delta method applied to the ratio $g(w, q) = w/(2q)$ yields asymptotic normality of $\widehat{nABCD}$:

$$\sqrt{n}(\widehat{nABCD} - nABCD(F_1, F_2)) \xrightarrow{d} N(0, \sigma_{nABCD}^2), \quad (A1)$$

where σ_{nABCD}^2 depends on the asymptotic variances and covariance of the W_1 and IQR estimators. This result requires $nABCD > 0$ (i.e., $F_1 \neq F_2$); at the boundary $F_1 = F_2$, the parameter lies on the edge of the parameter space and standard asymptotics do not apply (see Section 5).

The bootstrap provides valid inference for the Wasserstein distance under mild conditions.¹³ The percentile bootstrap confidence interval achieves asymptotically correct coverage for non-degenerate distributions.

B R CODE FOR NABCD COMPUTATION

Listing 1 R function for computing $nABCD$ with bootstrap confidence intervals.

```
compute_nABCD <- function(x1, x2) {
  pooled <- c(x1, x2)
  iqr_pooled <- IQR(pooled)
  if (iqr_pooled == 0) return(NA)
  all_vals <- sort(unique(pooled))
  F1 <- ecdf(x1); F2 <- ecdf(x2)
  w1 <- sum(diff(all_vals) *
    abs(F1(all_vals[-length(all_vals)]) -
      F2(all_vals[-length(all_vals)])))
  w1 / (2 * iqr_pooled)
}

nABCD_bootstrap <- function(x1, x2,
  B = 2000, conf = 0.95) {
  obs <- compute_nABCD(x1, x2)
  boot_vals <- replicate(B, {
    b1 <- sample(x1, replace = TRUE)
    b2 <- sample(x2, replace = TRUE)
    compute_nABCD(b1, b2)
  })
  alpha <- 1 - conf
  ci <- quantile(boot_vals,
    c(alpha/2, 1 - alpha/2), na.rm = TRUE)
  list(estimate = obs,
    ci_lower = ci[1], ci_upper = ci[2])
}
```